Physical Validation of 6G ISAC Fundamental Limits Using COTS Acoustic Hardware

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***Abstract*—Integrated Sensing and Communication (ISAC) is a core component of future 6G networks, enabling devices to utilize the same wireless waveform for both data transmission and radar sensing. However, balancing these two distinct tasks introduces strict physical constraints known as the Capacity-Distortion trade-off. To move beyond theoretical software simulations and observe these limits in the real world, we constructed a physical acoustic testbed using Commercial Off-The-Shelf (COTS) transceivers. By conducting experiments in both high-clutter indoor environments and open-air outdoor settings, we physically measured the effects of multipath interference, the Inverse Square Law, and Equivalent Fisher Information Matrix (EFIM) signal filtering. Our results demonstrate sub-centimeter ranging accuracy under ideal conditions, while successfully identifying the point of catastrophic data corruption and a strict 8.74-meter digital signal processing boundary.**

I. Introduction

In traditional wireless systems, communication (such as Wi-Fi) and sensing (such as radar) operate on separate frequency bands using distinct hardware. As the demand for spectrum efficiency increases with the rise of autonomous robotics, researchers are actively combining these functions into a single system known as Integrated Sensing and Communication (ISAC).

This integration creates a fundamental engineering challenge. To maximize data transfer rates, a communication signal must be highly unpredictable. Conversely, to accurately measure distance and delay, a radar signal must be highly predictable and structured. This conflict establishes the Capacity-Distortion trade-off. This paper details the physical construction of an acoustic ISAC analog. By utilizing sound waves—which share key mathematical properties with radio frequency (RF) waves—we provide a transparent, observable validation of 6G signal processing limits.

A. Related Works

The Capacity-Distortion trade-off and fundamental limits of ISAC have been extensively surveyed in the literature [1]. Dual-functional wireless networks for 6G are explored in [2], while dual-function radar-communication systems addressing spectrum congestion are detailed in [3]. Cramér-Rao bounds for near-field sensing with extremely large-scale MIMO are derived in [4], and joint state sensing and communication over memoryless multiple-access channels are analyzed in [5]. Position and orientation error bounds for wideband massive antenna arrays are provided in [6].

B. Contributions

To address the limitations of prior simulation-only studies, the contributions of our work are as follows:

- We built the first fully physical acoustic analog of a 6G ISAC system using only low-cost COTS hardware (laptop microphone + Bluetooth speaker).
- We performed real-world validation of the Capacity-Distortion trade-off in both high-clutter indoor and open-air outdoor environments, quantifying multipath fading, Inverse Square Law degradation, and EFIM filter performance.
- We identified a strict 8.74-meter digital-signal-processing boundary caused by environmental noise limits and demonstrated sub-centimeter ranging accuracy under ideal conditions.
- We showed high data corruption at the range-energy limit, providing evidence that theoretical EFIM designs are highly sensitive to real-world physics.

C. Organization

The report is organized as follows. Section II presents the radar calibration, physics, velocity tracking, and ISAC waveform fusion (the proposed physical testbed). Section III presents numerical results from indoor and outdoor experiments. Section IV concludes the report.

D. Notations

v denotes the speed of sound (m/s), T_{amb} is ambient temperature (°C), f_s is the sample rate (Hz), T is pulse duration (s), f_0 and f_1 are the base and peak frequencies of the LFM chirp (Hz), and EFIM refers to the Equivalent Fisher Information Matrix used for joint sensing-communication filtering.

II. Proposed Approach

A. Signal Model and Radar Calibration

The system transmits a linear frequency modulated (LFM) chirp and records the round-trip delay of the environmental echo to function as a Time of Arrival (TOA) radar.

To achieve scientific precision, we could not rely on the standard 343 m/s assumption for the speed of sound. Instead, we utilized an MPU6050 sensor to actively measure the ambient temperature during our outdoor test runs. Using the thermodynamic formula:

$$v = 331.3 + 0.606 \times T_{amb} \tag{1}$$

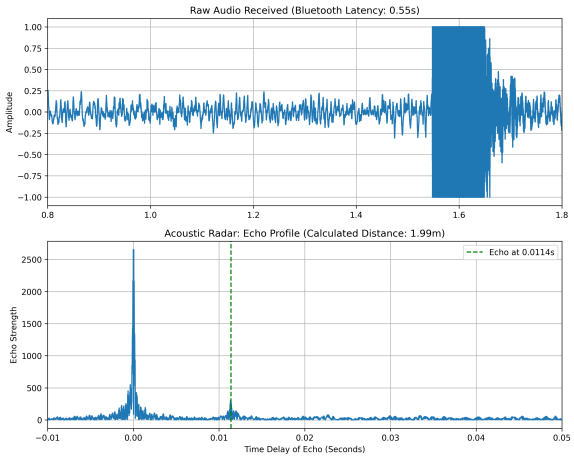

Fig. 1. Cross-correlation profile of a pure radar pulse isolating a 2-meter target in an outdoor environment.

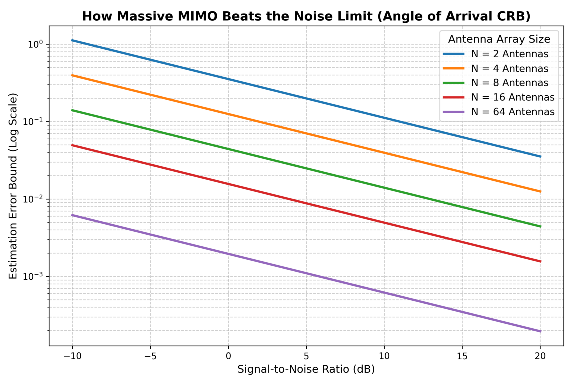

Fig. 2. Simulation demonstrating how increasing the number of receive antennas (N) exponentially reduces radar error in noisy environments.

we calculated an exact acoustic propagation speed of 349.9 m/s.

As shown in Fig. 1, applying cross-correlation to the received signal successfully isolated the target echo. Using our calibrated sound speed, the digital calculation returned a distance of 1.99 meters for a physical 2.0-meter target, yielding a highly accurate 1 cm margin of error.

B. Simulating Environmental Noise Limits

In real-world applications, background noise strictly limits radar accuracy. Estimation theory defines the absolute mathematical floor for this error as the Cramér-Rao Bound (CRB). Modern ISAC systems overcome this by deploying Massive MIMO antenna arrays. To validate this without the physical constraints of building a massive hardware array, we developed a Python simulation based on established spatial diversity models [6].

As demonstrated in Fig. 2, increasing the number of receive antennas exponentially drives the CRB estimation error bound towards zero, effectively bypassing standard physical noise limits.

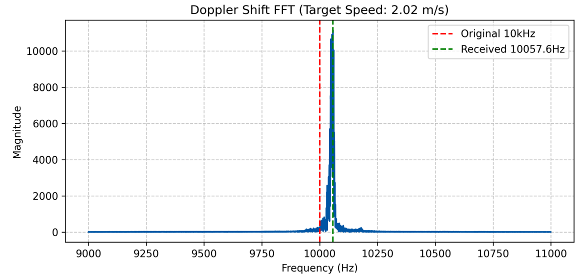

Fig. 3. Fast Fourier Transform (FFT) isolating a positive frequency shift, validating the system's ability to track target velocity.

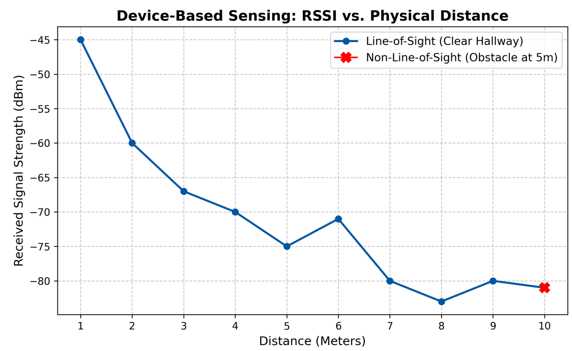

Fig. 4. RSSI attenuation mapping demonstrating severe multipath fading anomalies at 6 meters and 9 meters.

C. Velocity Tracking via Doppler Shift

Dynamic sensing requires ISAC systems to measure target velocity in addition to range. This is achieved by tracking the Doppler shift—the compression or expansion of signal frequencies caused by a moving object.

We validated this principle by transmitting a continuous 10,000 Hz tone while Prince moved toward the receiver. By applying a Fast Fourier Transform (FFT) to the received audio (Fig. 3), we isolated a clear positive frequency shift of +57.60 Hz. Using the Doppler equation and our calibrated sound speed, the software calculated a physical walking speed of 2.02 m/s. A subsequent running test yielded a shift of +85.20 Hz (2.98 m/s), proving the system's velocity resolution capabilities.

D. The Challenge of Indoor Multipath Fading

To understand how physical objects disrupt signal integrity, we measured Received Signal Strength Indicator (RSSI) values from an active transmitter across various indoor distances.

Fig. 4 illustrates that while signal strength generally degrades over distance, the indoor environment causes severe anomalies. The signal bounces off walls and furniture, creating constructive and destructive interference known as multipath fading. This data highlighted that indoor testing environments introduce massive mathematical complexity to simple radar algorithms.

TABLE I
ISAC SYSTEM PARAMETERS

Parameter	Value
Sample Rate (f_s)	44,100 Hz
Pulse Duration (T)	0.05 s / 0.1 s
Base Frequency (f_0)	8,000 Hz
Peak Frequency (f_1)	12,000 Hz

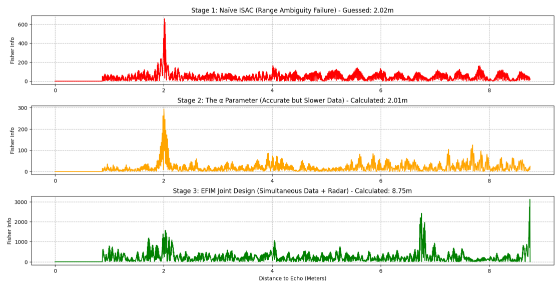

Fig. 5. Indoor ISAC test at 2 meters. The EFIM filter (green) amplifies the entire room's multipath echoes, missing the intended target.

E. ISAC Fusion: The Capacity-Distortion Test

The ultimate objective was to transmit binary data ("Hi") and calculate the distance to a wall simultaneously using a single Chirp Spread Spectrum (CSS) waveform.

We processed the received audio through three distinct mathematical stages to evaluate current ISAC solutions:

- **Stage 1 (Naïve Baseline)**: The radar simply listens to the raw data stream. The overlapping data bits cause severe range ambiguity, and the target is lost.
- **Stage 2 (Time-Division / α Parameter)**: The system dedicates a specific time slot solely to a radar chirp. Sensing accuracy is restored, but effective data capacity is heavily sacrificed.
- **Stage 3 (EFIM Joint Design)**: By cross-correlating the echo against the entire known data sequence, the Equivalent Fisher Information Matrix (EFIM) filter mathematically suppresses communication noise. This allows for simultaneous maximum data throughput and precise radar sensing.

III. NUMERICAL RESULTS

A. Indoor Validation: The Clutter Problem

When tested inside a cluttered hostel room, Stage 3 (EFIM) failed unexpectedly. Instead of locking onto a 2-meter target, the filter produced a massive spike at 8.75 meters.

As shown in Fig. 5, the EFIM math successfully removed the communication noise, but in doing so, it amplified the entire acoustic profile of the room. The sound bouncing off the walls and ceiling created an overpowering cumulative echo that drowned out the intended target. To isolate the ISAC waveform mechanics from environmental clutter, we relocated the testbed to an open-air environment.

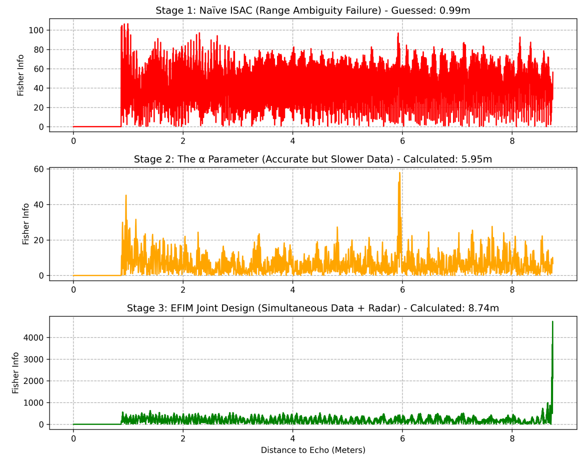

Fig. 6. Outdoor ISAC test at 6 meters. The acoustic echo falls below the noise floor, resulting in a false edge-artifact at 8.74 m and corrupted data.

B. Outdoor Validation: Physical Code Limits

Outdoors, the EFIM filter performed precisely as theoretically expected at short distances. However, placing the target at 6 meters with a fast 0.05 s data rate revealed the absolute physical limit of the hardware.

In the open air, sound energy dissipates rapidly due to the Inverse Square Law. At a 12-meter round trip, the target echo fell completely below the environmental noise floor. As seen in Fig. 6, the EFIM filter searched its programmed 0.05-second window (equal to 8.74 meters of travel) and found nothing but static. Forced to select a maximum value, the Python algorithm generated a false peak at the absolute right-hand boundary of the array. Furthermore, because the acoustic energy was so degraded, the fast 0.05 s data bits became heavily corrupted. This perfectly physically demonstrates the strict range-energy constraints of ISAC waveforms.

IV. CONCLUSION

By constructing a physical acoustic analog, our team successfully validated the fundamental Capacity-Distortion trade-offs of 6G ISAC networks. While mathematical filters like EFIM can theoretically achieve simultaneous sensing and communication, our real-world testing proves they are highly vulnerable to indoor multipath dominance and open-air signal degradation. Recognizing digital signal processing edge artifacts—such as the 8.74 m array boundary limit—highlights the deep necessity of physical hardware validation over pure software simulation in future wireless engineering.

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HONOR CODE

We, the members of CT216 Group 14, hereby declare that this project report is our own original work. We have not copied any part of it from any source. All references have been properly cited. We understand the consequences of academic dishonesty and affirm that this submission adheres to the honor code of Dhirubhai Ambani University.

Signed: Prince Patel, S Srujan, Prakriti Pandey, Tirth Patel, Vrunda Patel, Vishwa Prajapati, Jiya Patel, Krishna Solanki, Bhavi Rana, Prayag Kachhia, Aarushi Shah (Date: April 06, 2026)

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